



Review

Monitoring and adapting endurance training on the basis of heart rate variability monitored by wearable technologies: A systematic review with meta-analysis



Peter Dükling^{a,*}, Christoph Zinner^b, Khaled Trabelsi^c, Jennifer L. Reed^{d,e,f}, Hans-Christer Holmberg^{g,h}, Philipp Kunz^a, Billy Sperlich^a

^a Integrative and Experimental Exercise Science, Department of Sport Science, University of Würzburg, Germany

^b University of Applied Sciences for Police and Administration of Hesse, Germany

^c Education, Motricité, Sport et Santé, EM2S, LR19JS01, High Institute of Sport and Physical Education of Sfax, University of Sfax, Tunisia

^d Exercise Physiology and Cardiovascular Health Laboratory, Division of Cardiac Prevention and Rehabilitation, University of Ottawa Heart Institute, Canada

^e School of Human Kinetics, Faculty of Health Sciences, University of Ottawa, Canada

^f School of Epidemiology and Public Health, Faculty of Medicine, University of Ottawa, Canada

^g Department of Physiology and Pharmacology, Biomedicum C5, Karolinska Institutet, Sweden

^h Department of Health Sciences, Luleå University of Technology, Luleå, Sweden

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ABSTRACT

Objectives: To systematically perform a meta-analysis of the scientific literature to determine whether the outcomes of endurance training based on heart rate variability (HRV) are more favorable than those of predefined training.

Design: Systematic review and meta-analysis.

Methods: PubMed and Web of Science were searched systematically in March of 2020 using keywords related to endurance, the ANS, and training. To compare the outcomes of HRV-guided and predefined training, Hedges' g effect size and associated 95% confidence intervals were calculated.

Results: A total of 8 studies (198 participants) were identified comprising 9 interventions involving a variety of approaches. Compared to predefined training, most HRV-guided interventions included fewer moderate- and/or high-intensity training sessions. Fixed effects meta-analysis revealed a significant medium-sized positive effect of HRV-guided training on submaximal physiological parameters ($g = 0.296$, 95% CI 0.031 to 0.562, $p = 0.028$), but its effects on performance ($g = 0.079$, 95% CI -0.050 to 0.393, $p = 0.597$) and $\dot{V}O_{2peak}$ ($g = 0.171$, 95% CI -0.213 to 0.371, $p = 0.130$) were small and not statistically significant. Moreover, with regards to performance, HRV-guided training was associated with fewer non-responders and more positive responders. **Conclusions:** In comparison to predefined training, HRV-guided endurance training had a medium-sized effect on submaximal physiological parameters, but only a small and non-significant influence on performance and $\dot{V}O_{2peak}$. There were fewer non-responders regarding performance with HRV-based training.

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1. Introduction

Changes in cardiorespiratory fitness and endurance performance in response to different predefined training programs designed a priori for a specific period of time (e.g., 4 weeks) vary widely – from positive, none or even negative adaptations.^{1–4} Potential explanations for such variations include differences in age, sex, genetic predisposition and the type of training, as discussed in the scientific literature.^{2,5–8} Of these factors, only the type of training can be altered. Identifying the

optimal approach to training can be challenging for coaches and/or athletes who are most often responsible for making this decision.

Coaches may be aided by appropriate assessment and interpretation of autonomic cardiac regulation, which is known to play an important role in adaptations to endurance training.⁹ In several different populations (e.g., athletes and obese individuals), adaptation of training to heart rate variability (HRV) mediated by the vagal nerve enhances improvements in peak ($\dot{V}O_{2peak}$) and/or maximal ($\dot{V}O_{2max}$) oxygen consumption and endurance performance (e.g., 5-km running time).^{9–12} Consequently, it has been proposed that endurance training based on daily alterations in HRV may reduce variation in physiological adaptations and potentially improve performance more effectively.⁹ In the case of runners¹³ and other endurance athletes,¹⁴ recent systematic

* Corresponding author.

E-mail address: peter.dukling@uni-wuerzburg.de (P. Dükling).

reviews comparing HRV-based and predefined training concluded that the former leads to more pronounced improvement in certain parameters related to performance and physiological variables (e.g., running velocity at the first or second lactate threshold).¹³

Technological developments, particularly in wearable sensor technology (wearables) and associated algorithms, have greatly facilitated the assessment and interpretation of HRV.^{3,15} These wearables are lightweight devices worn in, on, or close to the body and designed to monitor, analyze, and interpret a variety of parameters, as well as transmit and/or receive data from other devices and/or cloud services, often providing direct feedback to the user.¹⁶

The aim of this systematic review with meta-analysis was to identify relevant scientific literature to compare the various effects of predefined training with training guided by variations in HRV regarding performance or physiological parameters.

2. Methods

Inclusion criteria

Our aim was to include all peer-reviewed publications that compare the outcomes (e.g., $\dot{V}O_{2peak}$ or $\dot{V}O_{2max}$) of predefined programs of endurance training with those based on variations in autonomic cardiac regulation.

Study design

To be eligible for inclusion, the study had to be experimental (e.g., randomized controlled trials, pre/post design, and quasi-experimental investigations) and include a control group that performed predefined endurance training.

Study populations

All investigations involving healthy endurance athletes, regardless of age, sex, gender, or level of performance, were included.

Interventions

To be eligible for consideration, the interventions were required to have the following characteristics, with appropriate reporting:

- Predominant involvement of endurance training
- Continuous adjustment of training by the intervention group on the basis of alterations in HRV
- Inclusion of a control group that performed a predefined program of training
- Utilization of wearable sensors
- Either null hypothesis significance testing (NHST) and/or magnitude-based inference (MBI) statistical analysis

With MBI statistical analyses, the potential benefit of the change observed was categorized as follows: most unlikely (<0.5%), very unlikely (0.5%–5%), unlikely (5%–25%), possible (25%–75%), likely (75%–95%), very likely (95%–99.5%), and most likely (>99.5%).¹⁷ When the 90% confidence limits overlap small positive or negative values, the benefit of the change is designated as unclear.¹⁷

Outcomes

For inclusion, maximal indicators of performance, such as time-to-volitional exhaustion during an appropriate test, in combination with peak or maximal (e.g., $\dot{V}O_{2peak}$ or $\dot{V}O_{2max}$) and submaximal (e.g., $\dot{V}O_{2submax}$, HR_{submax}) indicators of physiological condition had to be assessed.

Publication status and language

We limited our search to full-length original articles published in English in peer-reviewed journals (i.e., so-called “grey” literature, such as conference abstracts, dissertations, theses or reports, was not included).

Search strategy

Following the recommendations of the Preferred Reporting Items of Systematic Reviews and Meta-analysis (PRISMA),¹⁸ we carried out a systematic search of Ovid MEDLINE In-Process and Other Non-Indexed Citations, and Web of Science (with no restriction concerning publication date) in March of 2020 to identify articles of potential relevance using the following search profile: HRV OR heart rate variability OR autonomic nervous system AND guided OR training OR biofeedback AND endurance OR running OR exercise OR aerobic OR cycling OR swimming OR cross-country OR triathlon. In addition, the list of references in each of the articles identified in this manner was screened in attempt to discover any other articles of relevance that might have been missed using this search profile.

Selection of articles

After incorporating potentially relevant articles into EndNote X9 (Clarivate Analytics, Philadelphia, PA, USA) and eliminating duplicates, one author (PD) screened the titles and abstracts on the basis of our inclusion criteria and a second author (BS) independently confirmed those assessments. Thereafter, the full texts of the relevant articles were reviewed in detail for eligibility by these individuals (who were not blinded to the journals or authors involved). Cases in which there was disagreement were discussed, with or without input from a third author (CZ), until consensus was reached.

Data extraction and analysis

From each article thus identified, the first author extracted the details concerning publication (authors, journal, date); the participants (mean age, number, sex); the intervention (training mode, duration, wearables employed for monitoring HRV and related parameters, criteria for the modification of training); and the physiological and performance-related parameters evaluated (e.g., running performance, peak oxygen uptake, running velocity at lactate thresholds). A second author (BS) confirmed that the data were extracted correctly.

Utilizing the commercially available Comprehensive Meta-Analysis software (CMA version 3.0, Biostat, Englewood, NJ, USA), meta-comparisons of the intervention and control groups before and after training, based on mean values and standard deviations, together with sample sizes and pre-post correlation values, were performed. When pre-post correlations were not reported, we employed the conservative estimate of $r = 0.5$ recommended by Higgins and colleagues.¹⁹

Hedges' g value was used as the measure of effect size (ES), with values of 0.8, 0.5, and ≤ 0.2 being considered large, medium and small, respectively. In addition, point estimates of the effect size and 95% confidence intervals were illustrated with forest plots. A positive ES indicated that the effect of HRV-based training was superior to that of pre-defined training.

Statistical heterogeneity was assessed with the Q^{20} and I^2 ²¹ statistics. Due to the lack of substantial heterogeneity between studies, we applied a fixed-effects model for all comparisons.²¹

Potential publication bias was explored using visual evaluation of potential funnel plot asymmetry, the Egger's linear regression test,²² the Begg and Mazumdar's rank correlation test (Kendall's S statistic $P-Q$)²³ and the Duval and Tweedie's trim-and-fill test.²⁴

The stability of the pooled ES for each individual study was assessed by sensitivity analysis approach, i.e., computation of the impact of removing that particular study from the analysis (leave-one-out). To help further ensure stable and reliable results, a cumulative meta-analysis was conducted. For all analysis, $p < 0.05$ was considered to be statistically significant.

Assessment of methodological quality

The quality of the methodology employed in each of the articles included was assessed using the scale of the Physiotherapy Evidence Database (PEDro) (described in detail elsewhere).^{25,26} In brief, ten different aspects of each intervention (e.g., blinding of subjects and assessors, the extent to which the subjects in the experimental and

control groups differed prior to the intervention) were assessed and one point assigned for each positive assessment, resulting in a highest possible score of 10 points.^{25,26} Two of the authors (PD and CZ) assessed the risk of bias independently, resolving any disagreements through discussion, with or without the involvement of a third author (BS).

3. Results

Study characteristics

The total number of records identified and examined are shown in Fig. 1. A total of 5169 articles were identified in Web of Science, 5652 in Pubmed. Of these articles, eight met our inclusion criteria for analysis. All of these were published between 2007 and 2019 and received PEDro scores of 6–8 (the exact PEDro score of each article can be found in Table A).

Table 1 presents key aspects of the interventions analyzed, including the predefined program of training, methodology used to monitor HRV, and alteration of training by the intervention group in response to changes in HRV.

In one case, two intervention groups were involved,²⁹ i.e., the 8 studies involved a total of 9 interventions.

Altogether, 198 participants (55 women, 128 men and 15 participants in one study whose sex was not reported) performed training interventions ranging from 15 days²⁷ to four⁵ or eight weeks^{9,28–32} in

duration. The $\dot{V}O_{2peak}$ values for the male and female participants ranged from 49 to 68 ml·kg⁻¹·min⁻¹,^{5,9,29–32} and 35–59 ml·kg⁻¹·min⁻¹,^{27–29} respectively (One article²⁸ did not report this value prior to the intervention). Three interventions consisted of running only^{5,9,28}; another involved running approximately 60% of the training time (with ~13% cycling and ~27% other)²⁹; one running in combination with five sessions of strength training during the 8-week intervention³⁰; two cycling³²; and one hypoxic “live-high-train-low” Nordic skiing.²⁷

In 7 of the interventions,^{5,9,27–30} null-hypothesis testing alone was applied to assess intra- and/or intergroup differences, while the remaining two used magnitude-based inference statistics.^{31,32}

During 8 interventions HRV was assessed in the morning^{5,9,27,29–32} and in the afternoon/evening in the remaining intervention.²⁸ The parameters related to HRV that were assessed varied: in association with five interventions the root mean sum of squared differences (rMSSD) was determined^{9,28,30–32}; one focused on high-frequency power⁵; one the standard deviation associated with the instantaneous beat-to-beat variability in the R-R interval (obtained from Poincaré plots)²⁹; and the other spectral power analysis.^{27,29}

The periods used for determination of the baseline HRV for comparison with the daily values observed during the intervention were one,²⁷ three,³⁰ seven^{9,31,32} and 10 days.^{5,28,29}

Training parameters

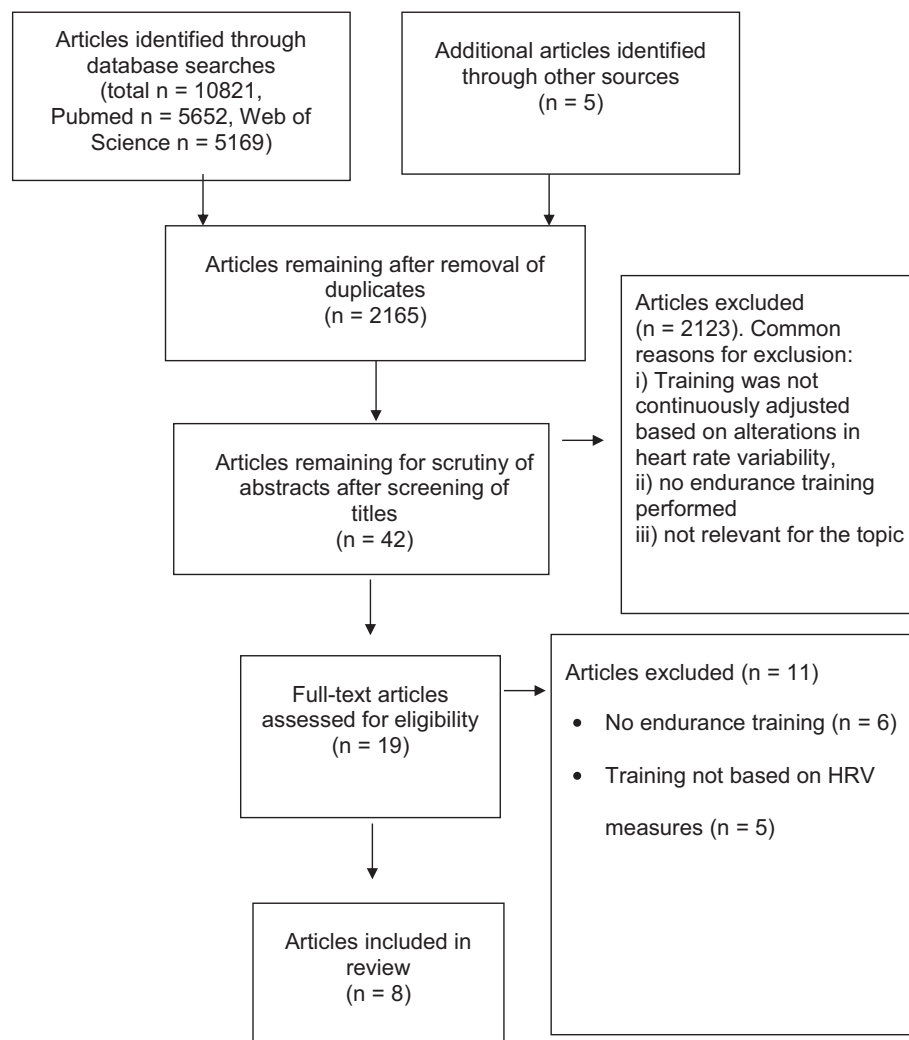


Fig. 1. Process of study selection, from initial identification as potentially relevant to inclusion.

Table 1
Characterization of the studies analyzed.
(Table adapted from: Düring, Zinner, Reed, Holmberg, & Sperlich¹³).

Article	N (male, female)	Participants age (yrs)	Level of performance	Sport	Pre-defined training characteristics	Technology employed for monitoring HRV	Measurement employed as a basis for HRV-guided training	Modification of training
(Javaloyes, Sarabia, Lamberts, & Moya-Ramon, 2018) ³¹	17 (17,0)	39.2 ± 5.3 (HRV-based)	$\dot{V}O_{2\max}$: 55.0 ± 7.58 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (HRV-based) 52.2 ± 6.5 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (Predefined)	Cycling	8-week intervention period Maintenance of total training volume Increase of training intensity (not specified any further) 4 x (30 min at VT2) 4 x (4 × 8 > VT2) 6 x (40 min between VT1 & 2) 20–32 x (between 120–180 min < VT1)	Polar H7 Chest strap (Polar Electro Oy, Kempele, Finland)	rMSSD determined every morning 90 s in supine position (first 30 s discarded) 7d moving window	No >2 consecutive days of rest No >2 consecutive sessions of moderate- or high-intensity training If rMSSD was less than the smallest worthwhile change, training intensity changed from moderate or high-to-low or rest
(Javaloyes, Sarabia, Lamberts, Plews, & Moya-Ramon, 2019) ³²	15 (n.r.)	28.1 ± 13.2 (HRV-based)	$\dot{V}O_{2\max}$: 58.9 ± 5.6 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (HRV-based) 59.0 ± 6.2 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (Predefined)	Cycling	8-week intervention period Maintenance of total training volume 4 training blocks: 3 high-intensity (consisting of 4 sessions (≥VT2) and HIITs (>VT2) per week) followed by a block of low-intensity (<VT1) training (4 sessions and 1 HIIT per week)	Smartphone flashlight and camera were used to assess pulse-rate variability at the fingertip using the smartphone application HRV4Training (www.hrv4training.com)	rMSSD was determined every morning 90 s in supine position 7d moving window	No >2 consecutive days of rest No >2 consecutive sessions of high-intensity training If rMSSD was below the smallest worthwhile change, training intensity changed from high to low or rest.
(Schmitt, Willis, Fardel, Coulmy, & Millet, 2018) ²⁷	24 (19, 5)	22.9 ± 4.3 (HRV-based, male), 20.5 ± 0.7 (female)	$\dot{V}O_{2\max}$: 67.8 ± 3.6 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (male) 58.9 ± 2.5 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (female)	Nordic skiing	HRV- based: 15 days of “live-high-train-low” sleeping with 15% O_2 in the air Hypoxic predefined: sleeping with 15% O_2 in the air Normoxic predefined All three groups trained in a similar manner as recommended by the national coach Training intensities 1) < VT1 2) VT1 – VT2 3) At VT2 4) For strength and speed sessions	Ambit 2 Peak (Suunto Oy, Vantaa, Finland)	HRV analyzed every morning RR intervals between the 3rd and 8th min supine, and between the 9th and 14th min standing. Spectral Power analysis with fast Fourier transformation (high-frequency power band 0.15–0.40 Hz; low-frequency power band 0.04–0.15 Hz)	Maintained or increased HF induced a higher TL than on the previous day. A ≥ 30% decrease in HF led to lower TL. Two consecutive days of reduced HF led to a day of rest. Following a day of rest, a session of low-intensity and medium-volume exercise was prescribed.
(Vesterinen et al., 2016) ⁹	40 (20,20)	34 ± 8 (female) 35 ± 7 (male)	$\dot{V}O_{2\text{peak}}$: 49 ± 4 (female) 56 ± 5 (male)	Running	38 training sessions 8 weeks 16 at low-intensity (below LT1) 16 at moderate or high-intensity Periodization: 3 weeks of intense training, 1 week of easy training, progressively increasing intensity 1 session of activity other than running each week	Omegawave Pro Mobile System (Omegawave Ltd., Helsinki, Finland)	rMSSD determined every morning 4-min supine position 7d moving window	4-week baseline period immediately before the intervention Smallest individual worthwhile changes calculated from the baseline data If 7d moving- window rMSSD < smallest worthwhile change: low-intensity training. If 7d moving- window rMSSD > smallest worthwhile change: high- or moderate-intensity training
(Kiviniemi et al., 2010) ²⁹	28 (14, 14)	35 ± 4 (HRV-based, male) 37 ± 3	$\dot{V}O_{2\text{peak}}$: Men: 50 ± 6 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ Women: 37 ± 3	Running	8 weeks training intervention 2 × 40 min at ~70% $HR_{\max}$ 3 × 30 min at ~85% $HR_{\max}$ (+5 min warm-up and cool-down)	Polar RS800 (Polar Electro Oy, Kempele, Finland)	SD1 determined every morning 3 min standing	Individual baseline data from the preceding 10 days (moving window) If no significant decrease in SD1

(continued on next page)

Table 1 (continued)

Article	N (male, female)	Participants age (yrs)	Level of performance	Sport	Pre-defined training characteristics	Technology employed for monitoring HRV	Measurement employed as a basis for HRV-guided training	Modification of training
(Kiviniemi et al., 2010) ²⁹	10 (0,10)	(Predefined, male) 33 ± 4 (HRV-based, female) 34 ± 4 (Predefined, female)	35 ± 5 ml·kg ⁻¹ ·min ⁻¹	Running	8 weeks training intervention 2 × 40 min at ~70% HR _{max} 3 × 30 min at ~85% HR _{max} (+5 min warm-up and cool-down)	Polar RS800 (Polar Electro OY, Kempele, Finland)	SD1 determined every morning 3-min standing	on any given day: vigorous exercise Moderate exercise after 2 sessions of vigorous exercise Significant decrease in SD1 on any given day: moderate intensity exercise Significant decrease in SD1 on 2 consecutive days: rest for at most 2 consecutive days, followed by moderate-intensity exercise Moderate exercise after 2 sessions of vigorous exercise
(Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007) ⁵	26 (26, 0)	31 ± 6 (HRV-based) 32 ± 5 (Predefined)	VO ₂ peak: 37 ± 5 ml·kg ⁻¹ ·min ⁻¹	Running	4 weeks training intervention period Day 1: Low-intensity exercise; Day 2: High-intensity; Day 3: High-intensity; Days 4–6: Repeat Days 1–3; Day 7: Rest High-intensity: 30 min at 85% HR _{max} (+5 min warm-up and cool-down at 65% HR _{max}) Low-intensity: 40 min at 65% HR _{max}	Polar S810i (Polar Electro OY, Kempele, Finland)	High-frequency power analysis (HF _{power} ; 0.15–0.40 Hz) of the spectral component of HRV determined every morning 5-min standing	Individual baseline data collected on the preceding 10 days (moving window) If HF _{power} increased significantly on any given day: 2 days of high-intensity exercise followed by low-intensity exercise If HF _{power} decreased significantly on any given day: low-intensity training If HF _{power} decreased significantly for 2 consecutive days: rest for at most 2 consecutive days, followed by low-intensity training Rest after 9 consecutive days of training
(Nuutila, Nikander, Polomoshnov, Laukkanen, & Häkkinen, 2017) ³⁰	32 (32,0)	29 ± 4 (HRV-based) 31 ± 5 years (Predefined)	VO ₂ peak: 54.2 ± 4.1 (Predefined) 53.6 ± 4.2 (HRV-based)	Running	8 weeks training intervention Low-intensity sessions: <Individual aerobic threshold Duration: self- selected between 30 and 90 min High-intensity interval sessions: 4 × 4 min at 90–95% HR _{max} (3-min of recovery); 3x10x30 sec at 95% v _{max} (3 min of recovery between sets)	Garmin 920 XT (Garmin Ltd., Schaffhausen, Switzerland)	"Quick recovery test" based on rMSSD (Firstbeat Technologies OY, Jyväskylä, Finland) Supine position 3-min data collection	Training was scheduled in blocks In-between these blocks, HRV data were used to modify training If HRV < baseline: Low-intensity training If HRV > baseline: The next block could be started

(da Silva, Ferraro, Adamo, & Machado, 2017) ²⁸	36 (0, 36)	27.7 ± 3.6 (Predefined) 25.8 ± 3.1 (HRV-based)	Not reported	Running	3 training sessions per week A total of 12 sessions of moderate intensity exercise (6 × 30 min at 75% v_{max} ; 6 × 40 min at 75% v_{max}); 12 sessions vigorous intensity exercise at 100% v_{peak} (interval series of total duration comparable to that of the moderate-intensity training) (8 weeks)	Polar RS800cx (Polar Electro OY, Kempele, Finland)	rMSSD measured in the afternoon/evening	Baseline: 10 days moving window If rMSSD < baseline: moderately intense exercise If rMSSD > 1SD above baseline: High-intensity exercise

Abbrev.: HRV, heart rate variability; HR_{max}, individual maximal heart rate; HF_{power}, high frequency power; LT1, first lactate threshold; rMSSD, root mean sum of squared differences; SD, standard deviation; v_{max} , maximal running velocity; $\dot{V}O_{2peak}$, individual maximal systemic oxygen uptake; VT1 2, first and second ventilatory threshold; Wk, week, n.r., not reported.

In 5 of the interventions, modifications in training based on alterations in HRV resulted in fewer moderate- and/or high-intensity sessions of exercise than with a predefined program of training^{5,9,28,29,31}; a greater number of moderate-intensity sessions in one intervention²⁹; and no difference in three cases^{27,30,32} (Table 2).

Parameters used as indicators of performance

Performance was assessed on the basis of peak power output, the results of a 40-minute time-trial (40TT), time required to run 3000 m and/or 5000 m, maximal running velocity, and time required to attain this velocity in 7 of the studies,^{5,9,28–32} which we have grouped together. Table 2 summarizes the pre- and post-values for these parameters.

The 14 ESs in combination demonstrated a small positive, but statistically insignificant effect (ES = 0.171, SE = 0.113, 95% CI –0.050 to 0.393, Z-value = 1.514, p = 0.130; Fig. 2) of HRV-guided training on performance parameters. No statistically significant heterogeneity was detected (Q = 8.918, df = 13, p = 0.779; I^2 = 0.000%).

Publication bias

Visual inspection of the funnel plot (Fig. 3) showed no evidence of publication bias concerning parameters related to performance, a conclusion confirmed by Begg and Mazumdar's rank correlation test (Kendall's S statistic P – Q = 19.00; tau without continuity correction = 0.209, z = 1.04, p = 0.149; tau with continuity correction = 0.198, z = 0.985, p = 0.162), as well as by Egger's linear regression test (intercept = 0.735, SE = 1.706, 95% CI –2.981 to 4.552, t = 0.431, df = 12, p = 0.337). The Duval and Tweedie's trim-and-fill test did not identify any missing study on either side of the plot.

Sensitivity and cumulative meta-analyses

The reliability and stability of our findings on parameters related to performance were confirmed by the sensitivity and cumulative meta-analyses.

Peak oxygen uptake

In 6 of the studies,^{5,27,29–32} $\dot{V}O_{2peak}$ was assessed. Table 3 summarizes the values obtained before and after the interventions.

The 9 ESs in combination revealed a small-positive, but statistically insignificant effect (ES = 0.079, SE = 0.149, 95% CI –0.213 to 0.371, Z-value = 0.529, p = 0.597; Fig. 2) of HRV-guided training on $\dot{V}O_{2peak}$, with no significant heterogeneity (Q = 3.227, df = 8, p = 0.919; I^2 = 0.000%).

Publication bias

A visual inspection of the funnel plot (Fig. 3) showed no evidence of publication bias regarding $\dot{V}O_{2peak}$; a conclusion confirmed by Begg and Mazumdar's rank correlation test (Kendall's S statistic P – Q = 14.00; tau without continuity correction = 3.389, z = 1.46, p = 0.07; tau with continuity correction = –0.361, z = 1.355, p = 0.088), as well as by Egger's linear regression test (intercept = 2.543, SE = 1.695, 95% CI –1.464 to 6.551, t = 1.501, df = 7, p = 0.09). The Duval and Tweedie's trim-and-fill test did not identify any missing study.

Sensitivity and cumulative meta-analyses

In summary, the reliability and stability of our findings concerning $\dot{V}O_{2peak}$ were confirmed by the sensitivity and cumulative meta-analyses.

Submaximal physiological parameters

For this meta-analysis, the data from the 6 studies^{5,9,27,30–32} (comprising 61 participants) that evaluated the effects of both HRV-guided and predefined training on submaximal physiological parameters before and after the intervention were pooled (Table 3). The effects on lactate thresholds 1 (LT1) and 2 (LT2) was utilized to evaluate by Vesterinen⁹ and Nuutila³⁰, while Javaloyes et al.^{31,32} evaluated the effects on ventilatory thresholds and the corresponding power output (i.e., WVT1, WVT2). Each parameter evaluated was analyzed independently.

Table 2

Differences between the HRV-based and predefined endurance training and their effects on performance parameters.

(Table adapted from: Düring, Zinner, Reed, Holmberg, & Sperlich¹³).

Study	Differences between the HRV-based and predefined training programs	Performance						Comparison of the effects of HRV-based and predefined training
		HRV			Predefined			
		PRE	POST	Change (%)	PRE	POST	Change (%)	Result of NHST and/or MBI
(Javaloyes, Sarabia, Lamberts, & Moya-Ramon, 2018) ³¹	Training time per week	PPO:	PPO:	PPO:	PPO:	PPO:	PPO:	PPO:
	HRV-based: 9 h 18 min ± 2 h 50 min	356.83 ± 39.74	374.28 ± 43.65	5.1 ± 4.5* Likely beneficial	346.75 ± 16.73	351.50 ± 17.01	1.4 n.s. Unclear	n.s. HRV-based: Possibly beneficial
	Predefined: 8 h 46 min ± 2 h 47 min							
(Javaloyes, Sarabia, Lamberts, Plews, & Moya-Ramon, 2019) ³²	Intensity distribution: (<VT1/ VT1 - VT2/> VT2):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):
	HRV-based: 66/24/10%	243.11 ± 41.73	260.78 ± 44.76	7.3 ± 4.5* Very likely beneficial	214.42 ± 32.36	223.13 ± 36.15	4.2 n.s. Possibly beneficial	n.s. HRV-based: Likely beneficial
	Predefined: 64/27/9%							
(Javaloyes, Sarabia, Lamberts, Plews, & Moya-Ramon, 2019) ³²	Predefined training involved significantly more time between VT1 & VT2							
	Training time per week: HRV-based: 11 h 6 min ± 3 h 4 min	PPO: 395 ± 39	PPO: 423 ± 28	PPO: 7 ± 5* Very likely beneficial	PPO: 388 ± 42	PPO: 407 ± 51	PPO: 4.9 n.s.	PPO: n.s. HRV-based:
	Predefined: 11 h 22 min ± 3 h 7 min						Possibly beneficial	Unclear
(Schmitt, Willis, Fardel, Coulmy, & Millet, 2018) ²⁷	n.s.; (MBI: unclear)							
	Intensity distribution: (<VT1/ VT1 - VT2/>VT2)							
	HRV-based: 49/39/12%	40TT: 261 ± 28	40TT: 280 ± 39	40TT: 6 ± 6* Likely beneficial	40TT: 262 ± 30	40TT: 264 ± 33	40TT: 0.7 n.s. Possibly trivial	40TT: n.s. HRV-based: Unclear
(Vesterinen et al., 2016) ⁹	Predefined: 54/33/13% n.s. (MBI: unclear)							
	TRIMP did not differ between groups							
(Kiviniemi et al., 2010) ²⁹	HRV-based: 3365 ± 425 a.u.	Roller-ski performance after tapering	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:
	Predefined hypoxic: 3481 ± 179 a.u.	n.r.	n.r.	-2.7 ± 3.6*	n.r.	n.r.	-2.5 ± 3.5 n.s.	n.r.
	Daily training by the HRV-based group was adjusted on 3.3 ± 2.3 of the 15 days							
(Kiviniemi et al., 2010) ²⁹	Sessions of moderate- and high-intensity:	3000 m (km·h ⁻¹):	3000 m (km·h ⁻¹):	3000 m (%)	3000 m (km·h ⁻¹):	3000 m (km·h ⁻¹):	3000 m (%)	3000 m (km·h ⁻¹):
	HRV-based: 13.2 ± 6.0*	15.4 ± 1.6	15.7 ± 1.5	2.1 ± 2.0*	15.0 ± 1.6	15.2 ± 1.5	1.1 ± 2.7 (n.s)	0.42 (small effect)
	Predefined: 17.7 ± 2.5							
(Kiviniemi et al., 2010) ²⁹	Effect size: 0.98 (large)							
(Kiviniemi et al., 2010) ²⁹	Number of moderate-intensity sessions per week:	Load _{max} (Watt):	Load _{max} (Watt):	Load _{max} (%):	Load _{max} (Watt):	Load _{max} (Watt):	Load _{max} (%):	Load _{max} (Watt):
	HRV-based:	Men: 270 ± 29	Men: 300 ± 25	Men: +11.1	Men: 275 ± 28	Men: 293 ± 35	Men: 6.5*	Men: p < 0.1
	Men: 2.7 ± 0.2*							
(Kiviniemi et al., 2010) ²⁹	Women: 2.5 ± 0.3*	Load _{max} (Watt):	Load _{max} (Watt):	Load _{max} (%):	Load _{max} (Watt):	Load _{max} (Watt):	Load _{max} (%):	Load _{max} (Watt):
	Predefined:	Women: 174 ± 28	Women: 189 ± 25	Women: +8.6	Women: 179 ± 32	Women: 198 ± 35	Women: 10.6*	Women: n.s.
	Men: 2.1 ± 0.1							
(Kiviniemi et al., 2010) ²⁹	Women: 2.2 ± 0.2	Load _{max} (Watt):	Load _{max} (Watt):	Load _{max} (%):	Load _{max} (Watt):	Load _{max} (Watt):	Load _{max} (%):	Load _{max} (Watt):
	Number of intense sessions of exercise per week:	Women: 177 ± 26	Women: 194 ± 23	+9.6	Women: 179 ± 32	Women: 198 ± 35	Women: 10.6*	n.s.
	HRV-based: 1.8 ± 0.3*							
(Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007) ⁵	Predefined: 2.8 ± 0.6							
	Number of high-intensity sessions of exercise per week:	v _{max} (km·h ⁻¹):	v _{max} (km·h ⁻¹):	v _{max} (%):	v _{max} (km·h ⁻¹):	v _{max} (km·h ⁻¹):	v _{max} (%):	v _{max} (km·h ⁻¹)
	HRV-based: 3*	15.5 ± 1.0	16.4 ± 1.0	+5.8 *	15.1 ± 1.3	15.7 ± 1.2 *	3.9 *	p < 0.05
(Nuutila, Nikander, Polomoshnov, Laukkanen, & Hakkinen, 2017) ³⁰	Predefined: 4							
(da Silva, Ferraro, Adamo,	Number of training sessions per week	Total time required for	Total time required for	Total time required for	Total time required for	Total time required for	Total time required for	Total time required for
	HRV-based: 6.3 ± 1.4 (3 ± 3% in HR Zone 3)	3000 m (s)	3000 m (s)	3000 m (%)	3000 m (s)	3000 m (s)	3000 m (%)	3000 m
	Predefined: 6.1 ± 0.4 (4 ± 3% in HR Zone 3)	v _{max} (km·h ⁻¹):	v _{max} (km·h ⁻¹):	v _{max} (%)	v _{max} (km·h ⁻¹):	v _{max} (km·h ⁻¹):	v _{max} (%):	v _{max} (km·h ⁻¹):
(da Silva, Ferraro, Adamo,		17.6 ± 1.3	18.5 ± 1.2	+5.1*	18.0 ± 1.1	18.5 ± 1.2*	2.7 *	0.00 (trivial)
								v _{max} (km·h ⁻¹):
								0.95 (large effect)
(da Silva, Ferraro, Adamo,	Total relative number of	v _{max}	v _{max}	v _{max} (%39):	v _{max}	v _{max}	v _{max} (%)	v _{max} (km·h ⁻¹):

Table 2 (continued)

Study	Differences between the HRV-based and predefined training programs	Performance						Comparison of the effects of HRV-based and predefined training	
		HRV			Predefined				
		PRE	POST	Change (%)	PRE	POST	Change (%)		
& Machado, 2017) ²⁸	moderately intense sessions: HRV-based: 36.7 ± 5.5 * Predefined: 50 ± 0	(km·h ⁻¹):	(km·h ⁻¹):	+10.0 ± 7.3*	(km·h ⁻¹):	(km·h ⁻¹):	8.2 ± 4.7%*	0.29 (small effect)	
		10.9 ± 1.2	11.9 ± 0.9		11.0 ± 1.4	11.9 ± 1.4			
		t _{lim} at v _{max} (min):	t _{lim} at v _{max} (min):	t _{lim} at v _{max} (%):	t _{lim} at v _{max} (min):	t _{lim} at v _{max} (min):	t _{lim} at v _{max} (%):	t _{lim} at v _{max} (n.s.)	0.93 (moderate effect)
		5.1 ± 1.3	6.1 ± 1.7	+ 23.6 ± 31.9*	5.4 ± 1.7	4.8 ± 1.6	26.7 ± 31.2 (n.s.)		
		t5km (min):	t5km (min):	t5km (%):	t5km (min):	t5km (min):	t5km (%):	t5km:	
		36.3 ± 4.5	29.8 ± 2.4	-17.5 ± 5.6*	35.5 ± 5.0	30.5 ± 4.3	-14.0 ± 4.7 *	0.67 (moderate effect)	

Abbrev.: ES, effect size; 40TT, 40 min time trial; 3000 m, mean velocity in a 300 m run; Load_{max}, maximal load in a cycle ergometer test; v_{max}, maximal running velocity; t_{lim}, time run at maximal velocity; t5km, time to run 5000 m; n.r. = not reported; n.s. = not significant; * sign. different, a.u., arbitrary units; PPO, peak power output; n.s., not significant; MBI, magnitude based inferences;

The 10 ESs in combination demonstrated a statistically significant small-to-medium positive effect (ES = 0.296, SE = 0.135, 95% CI 0.031 to 0.562, Z-value = 2.191, $p = 0.028$; Fig. 2) of HRV-guided training on submaximal physiological parameters. The heterogeneity calculated ($Q = 6.424$, $df = 9$, $p = 0.697$; $I^2 = 0.000\%$) was insignificant.

Publication bias

Visual inspection of the funnel plot (Fig. 3), as well as the results of the Begg and Mazumdar's test (Kendall's S statistic $P-Q = 37.00$; tau without continuity correction = 0.822, $z = 3.309$, $p = 0.000$; tau with continuity correction = 0.8, $z = 3.219$, $p = 0.000$) and Egger's linear regression test (intercept = 4.379, SE = 1.291, 95% CI 1.402 to 7.357, $t = 3.391$, $df = 8$, $p = 0.005$) indicated publication bias with respect to the submaximal physiological parameters. Correction of this asymmetry using Duval and Tweedie "trim and fill" method yielded 4 potentially missing studies, and an adjusted ES of 0.458 (95% CI 0.241 to 0.675, $Q = 10.808$).

Sensitivity and cumulative meta-analyses

The sensitivity and cumulative meta-analyses confirmed the reliability and stability of these findings on submaximal physiological parameters.

Variations in performance

Four of the articles described the response to HRV-based and predefined training regarding performance during a 40TT and 3000-m running, as well as maximal running velocity.^{9,30–32}

In two cases fewer negative responders were present in the intervention group.^{31,32} Indeed, of the 16 participants in these two interventions, only two demonstrated a decline in performance (by approximately 1% and 20.5%) during the 40TT. In contrast, 5 of the 16 participants in the control groups who performed predefined training exhibited poorer 40TT performance (with declines of -2%, -5%,³² and -11.6%, -9.1% and -2.7%).³¹ In both of these studies, the average improvement in 40TT performance was more pronounced following HRV-based than predefined training.^{31,32} Another study found that the variation in 3000-m running performance was smaller following HRV-based (-1% to 6%) than predefined training (-4% to 8%).⁹ In three other studies, HRV-based training led to the greatest individual improvements in performance (i.e. in 40TT, maximal running velocity).^{30–32}

4. Discussion

This systematic review and meta-analysis compared the effects of predefined training with those of training guided by variations in HRV, as detected by wearables.

The major observations were as follows:

- 1) 5 of the 9 interventions involved fewer sessions of moderate- and/or high-intensity exercise than predefined training.
- 2) In comparison to predefined training, HRV-guided training exerted a small positive, but statistically insignificant effect on parameters related to performance (i.e., peak power output, performance in connection with a 40TT or 3000 m and/or 5000 m of running, maximal velocity, and time required to attain this velocity) and $\dot{V}O_{2peak}$.
- 3) In comparison to predefined training, HRV-guided training had a significant moderate-positive effect on submaximal physiological parameters.
- 4) With respect to performance, some participants demonstrated worsening, but more showed improvement following HRV-guided training.
- 5) Publications in this area to date have involved a wide variety of methodological approaches (e.g., different parameters related to HR, different training interventions). Similar research on, e.g., a variety of different study populations is needed.

Evaluation of the functional status of the autonomic nervous system (ANS) could reveal important aspects of physiological responses to training,^{33,34} since this system is the primary regulator of fundamental aspects of overall recovery, including delivery of oxygen and nutrients, removal of waste products, and thermoregulation by the cardiovascular system.³³

As described eloquently elsewhere,⁵ extensive vagal activity prior to training for 8 weeks (six 30–40 min sessions per week at 70–80% peak heart rate) improves aerobic power.¹¹ On the basis of other previous findings it has been proposed that the more pronounced enhancement in average performance following HRV-guided endurance training is due to more optimal timing of (more intensive) training sessions.^{5,31,32}

Our present meta-analysis indicates the primary involvement of improvements in submaximal physiological variables (moderate effect), with smaller improvements in $\dot{V}O_{2peak}$ (small effect) exerting less impact. However, additional investigations on the physiological mechanism(s) underlying these improvements in performance need to be carried out, especially since the methodology used in the articles reviewed varied considerably.

The only 4 studies^{9,30–32} that evaluated individual variations in the effects of training on performance (i.e., during a 40TT or 3000-m running, as well as maximal running velocity) demonstrated that in contrast to predefined training, HRV-guided training resulted in fewer participants with worsened performance, whereas a larger number showed more extensive improvement. Thus, for the individual HRV-guided training appears to produce results that are more favorable than those obtained with predefined training.

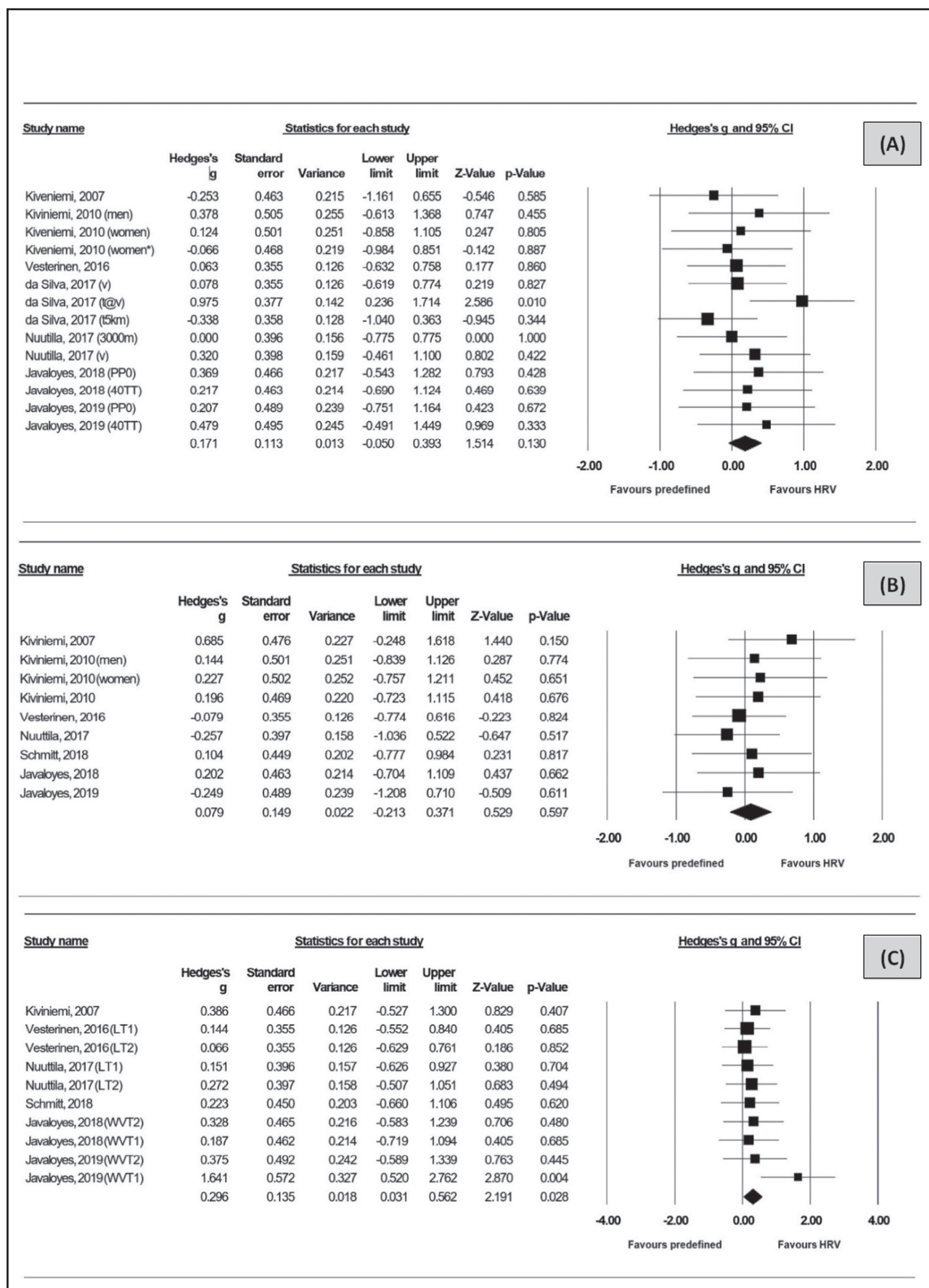


Fig. 2. Forest plots for the effects of HRV-guided endurance training compared to predefined training on performance parameters (A), $\dot{V}O_{2peak}$ (B) and submaximal parameters (C).

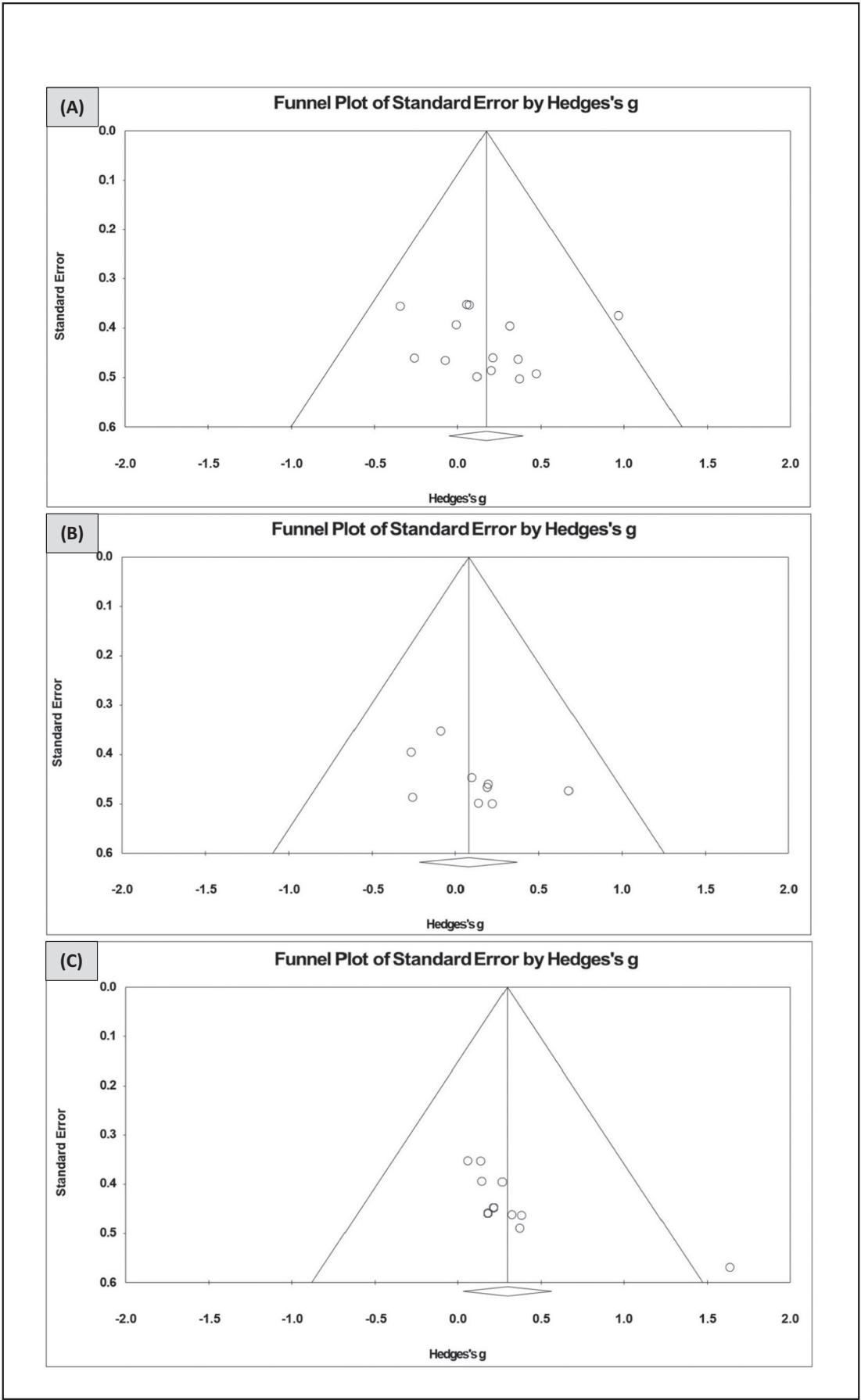


Fig. 3. Funnel plots of standard error by Hedge's g for performance parameters (A), VO2peak (B) (showing no evidence of publication bias) and submaximal physiological parameters (C).

Table 3
Physiological parameters following HRV-based and predefined endurance training.
(Table adapted from: Duku, Zinner, Reed, Holmberg, & Sperlich¹³.)

Study	Submaximal parameters									
	VO _{2peak} (ml · kg ⁻¹ · min ⁻¹)				Difference between HRV & Predefined		Difference between HRV & Predefined		Difference between HRV & Predefined	
	HRV		Predefined		Predefined		Predefined		Predefined	
	PRE (mean ± SD)	POST (%)	Change (%)	PRE (mean ± SD)	POST (mean ± SD)	Change (%)	PRE (mean ± SD)	POST (mean ± SD)	Change (%)	Difference between HRV & Predefined
(Javaloyes, Sarabia, Lamberts, & Moya-Ramon, 2018) ³¹	56.34 ± 7.58	55.8 ± 8.18	n.s. Unlikely beneficial	54.30 ± 7.81	52.13 ± 6.78	n.s. Very unlikely beneficial	WV2 (power output @ VT2): 275.00 ± 41.46	WV2 (power output @ VT2): 311.11 ± 37.73	WV2 (power output @ VT2): 281.25 ± 22.16	WV2 (power output @ VT2): n.s. HRV-based: Possibly beneficial
(Javaloyes, Sarabia, Lamberts, Plews, & Moya-Ramon, 2019) ³²	58.94 ± 5.62	61.01 ± 6.01	3 ± 3* Likely Beneficial	58.96 ± 6.23	62.65 ± 6.65	6.2 Likely Beneficial	WVT1 (power output @ VT1): 191.67 ± 27.95	WVT1 (power output @ VT1): 200.00 ± 25.01	WVT1 (power output @ VT1): 178.13 ± 28.15	WVT1 (power output @ VT1): n.s. Possibly beneficial
(Schmitt, Willis 2018) ²⁷	66.7 ± 5.9	69.2 ± 6.5	+3.8 ± 3.1*	63.7 ± 4.4	65.6 ± 4.9	3.0 ± 4.4 n.s.	WVT2: 288 ± 52	WVT2: 349 ± 30	WVT2: 323 ± 52	WVT2: n.s. HRV-based: likely trivial
(Vesterinen et al., 2016) ⁹	54.4 ± 6.2	56.4 ± 7.0	+3.7 ± 4.6*	53.0 ± 5.8	55.5 ± 5.8	+5.0 ± 5.2* n.s.	WVT1: 170 ± 37	WVT1: 234 ± 30	WVT1: 190 ± 42	WVT1: n.s. HRV-based: Unclear
(Kiviniemi et al., 2010) ²⁹	50 ± 6	54 ± 6	+8.0	50 ± 7	53 ± 7	+6.0*	VO ₂ VT ₂ : 55.8 ± 4.0	VO ₂ VT ₂ : 59.6 ± 5.9	VO ₂ VT ₂ : 57.1 ± 4.7	VO ₂ VT ₂ : n.s.
(Kiviniemi et al., 2010) ²⁹	37 ± 5	40 ± 5	+8.1	35 ± 5	37 ± 4	+5.7	LT1 (km·h ⁻¹): 11.0 ± 1.2	LT1 (km·h ⁻¹): 11.3 ± 1.2	LT1 (km·h ⁻¹): 10.7 ± 1.5	LT1 (km·h ⁻¹): 0.54 (moderate)
(Kiviniemi, Hautala, Kinnunen, & Tuuppo, 2007) ⁵	56 ± 4	60 ± 5	+7.1*	54 ± 4	55 ± 3	+1.9	LT2 (km·h ⁻¹): 13.4 ± 1.4	LT2 (km·h ⁻¹): 13.8 ± 1.3	LT2 (km·h ⁻¹): 13.1 ± 1.5	LT2 (km·h ⁻¹): 0.25 (small)
(Nuutila, Nikander, Polomoshnov, Laukkanen, & Hakkinen, 2017) ³⁰	53.6 ± 4.2	56.7 ± 3.4	+5.7*	52.2 ± 4.1	56.4 ± 4.7*	+8.0*	Men 50 ± 7	Men 53 ± 7	Men 57.1 ± 4.7	Men 0.25 (small)
(da Silva, Ferraro, Adamo, & Machado, 2017) ²⁸	n.f.	n.f.	n.f.	n.f.	n.f.	n.f.	Women 35 ± 5	Women 37 ± 5	Women 37.1 ± 4.7	Women 0.37 (small effect)

Abbrev.: NHST: Null hypothesis significance testing; LT1, first lactate threshold; LT2, second lactate threshold; MBI, magnitude-based inferences; WV1, Watt at first ventilatory threshold; WV2, Watt at second ventilatory threshold; $\dot{V}O_{2peak}$, individual maximal systemic oxygen uptake; ES, effect size; 3000 m, time to run 3000 m; n.r. = not reported; n.s. not significant.

Of course, at present this proposal is highly preliminary and a more detailed understanding of the processes involved is required. Studies which analyzed changes in performance at the individual level were simply counted and/or reported, but such “responder” counts are biased due to random intraindividual variability in the outcome(s) assessed before and after the intervention.³⁵ New approaches for estimating the proportion of responders in a population of interest that are less prone to bias involve the standard deviation for true individual responses, thereby directly incorporating information about the control group.³⁵ In addition, none of the investigations examined interindividual variations in the effects of training on physiological parameters (e.g., $\dot{V}O_{2peak}$, submaximal parameters) and such information could help elucidate the mechanisms underlying the more pronounced effects of HRV-guided training on performance.

With the growing availability of novel sensors and algorithms³⁶ that effectively monitor heart rate, it is becoming increasingly more feasible to base one's training on the variability in this parameter. Recreational and elite athletes who perform endurance training without a coach may benefit most in this respect.¹⁵ However, most of the consumer-oriented wearable sensors designed to monitor HRV have yet to be validated and consumers must always keep this limitation in mind.³⁶

The strengths of this systematic review and meta-analysis include our comprehensive search of the literature (including more than two scholarly databases, as recommended by the Cochrane Association guidelines and good practices for conducting systematic reviews), followed by careful appraisal of the quality of the studies included. Interestingly, the methodological approach in all of the publications we reviewed was of high quality, as indicated by their PEDro scores. In addition, there was no evidence of any significant heterogeneity across these studies, which further strengthens our conclusions.

With respect to the parameters related to performance (peak power output during a 40-min time-trial or 3000-m and/or 5000-m run, maximal velocity, and time required to attain that velocity) and $\dot{V}O_{2peak}$, there was no evidence of publication bias. However, this was not the case for submaximal physiological parameters. Our results must, therefore, be interpreted with caution.

We did not contact researchers in sports and exercise sciences for information on ongoing studies or manuscripts in press or available as pre-prints, and our search was limited to full-length original articles in English and published in peer-reviewed journals, which means that we might have missed some important findings.

In four of the publications,^{27,30–32} the sample size required to reach significant conclusions was not calculated a priori (e.g., using G*Power software³⁷). Many other aspects of the methodologies involved (e.g., wearable sensors and the procedures utilized to measure different parameters related to HRV, alteration of training in response to variations in this parameter) varied between studies, which complicates comparison. This variation, along with the current availability of only relatively few studies, makes the determination of a “best-practice” approach impossible at present. However, it has been argued previously that derivation of rMSSD and SD1 from a Poincaré plot is the most useful approach for close monitoring of athletes.³⁸

It remains unclear how training should best be altered in response to a reduction in HRV. For instance, should training be paused completely or high-intensity training replaced with low-intensity sessions? In addition, none of the articles included strategies for recovery, e.g., immersion in cold water, massages or other measures designed to reduce fatigue following endurance exercise.³⁹ It is possible that the inclusion of such procedures would be more beneficial to performance and/or physiological variables than alterations in training.

In addition, factors other than training, including psychological stress related to work or academic studies, exert an impact on HRV³³ and, therefore, we cannot be certain what caused the changes in HRV in the interventions described here. In this context, characterization of

individual stress³ would be desirable, potentially leading to strategies for reducing this stress and thereby improving recovery and/or training.

5. Conclusions

Training that was adapted in response to changes in HRV, in comparison to predefined endurance training, resulted in a significantly greater medium improvement in submaximal physiological parameters, whereas the differences with respect to performance and $\dot{V}O_{2peak}$ were small and not statistically significant. At the same time, with respect to performance, HRV-based training tended to be associated with fewer negative and a larger proportion of positive responders. The scientific literature in this area is sparse and the studies that have been performed have used a variety of methodological approaches, rendering definitive conclusions impossible to draw at the present time.

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PD, CZ, KT, JR, HC, PK and BS have no conflicts of interest to declare.

Confirmation of ethical compliance

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